

BLUESTOCK MUTUAL FUND

ANALYTICS — CAPSTONE REPORT

End-to-End Data Analytics Platform for Indian Mutual Funds

Data Analytics Internship | Bluestock Fintech | June 2026

40 Fund Schemes | 10 AMCs | 2022 – 2026

Table of Contents

- 1. Executive Summary 3
- 2. Project Overview & Objectives 4
- 3. Data Sources & Description 5
- 4. ETL Pipeline & Data Architecture 6
- 5. Exploratory Data Analysis (EDA) 8
- 6. Fund Performance Analytics 11
- 7. Advanced Analytics & Risk Metrics 13
- 8. Dashboard Overview 15
- 9. Key Findings & Insights 16
- 10. Limitations & Recommendations 17

1. Executive Summary

This report presents a comprehensive data analytics solution developed for Bluestock Fintech as part of the Capstone Project I — Mutual Fund Analytics. The project delivers an end-to-end analytics pipeline covering data ingestion, cleaning, database design, exploratory analysis, performance metrics, advanced risk analytics, and interactive dashboards for 40 Indian mutual fund schemes across 10 Asset Management Companies (AMCs) spanning the period January 2022 to December 2025.

The platform processes over 64,000 daily NAV records, 32,000+ investor transactions, and 8,000+ benchmark index values to produce actionable insights for fund managers, investors, and analysts. Key deliverables include a SQLite star-schema database, 15+ analytical charts, a composite fund scorecard ranking all 40 schemes, and a 4-page Tableau Public dashboard.

Key Metrics at a Glance

40	10	64,320+	32,778	■81L Cr
Fund Schemes	AMCs Covered	NAV Records	Investor Txns	Total AUM
■31,002				
Cr	26.12 Cr	6.5%	13 Charts	4 Pages
Peak SIP (Dec 2025)	Total Folios	Risk-Free Rate	Visualisations	Dashboard

2. Project Overview & Objectives

The Bluestock MF Analytics Capstone Project was designed to build a production-grade data analytics platform that mirrors real-world fintech analytics workflows. The project was structured across 7 working days, each focusing on a distinct phase of the analytics lifecycle.

Project Objectives

- Design and implement a complete ETL pipeline for Indian mutual fund data
- Build a normalised SQLite star-schema database with 11 tables
- Perform comprehensive EDA with 15+ interactive visualisations
- Compute institutional-grade performance metrics: CAGR, Sharpe, Sortino, Alpha, Beta, Max Drawdown
- Conduct advanced risk analytics including Historical VaR, CVaR, and HHI concentration
- Develop an interactive 4-page Tableau Public dashboard
- Build a fund recommender system based on investor risk appetite

Technology Stack

Component	Technology	Purpose
Language	Python 3.11	Core development
Data Processing	pandas, numpy	ETL and transformation
Visualisation	Plotly, Seaborn, Matplotlib	Charts and graphs
Database	SQLite + SQLAlchemy	Star schema storage
Statistics	scipy.stats	OLS regression (Alpha/Beta)
Dashboard	Tableau Public	Interactive 4-page dashboard
Version Control	Git / GitHub	Code management
Live Data	mfapi.in API	Real-time NAV fetching

3. Data Sources & Description

The project utilises 10 structured CSV datasets provided by Bluestock Fintech, supplemented by live NAV data fetched from the public mfapi.in API. All data pertains to AMFI-registered mutual fund schemes in India.

#	Dataset	Rows	Columns	Key Fields
1	01_fund_master.csv	40	15	amfi_code, fund_house, category, risk_category
2	02_nav_history.csv	64,320*	3	amfi_code, date, nav
3	03_aum_by_fund_house.csv	90	5	date, fund_house, aum_crore
4	04_monthly_sip_inflows.csv	48	6	month, sip_inflow_crore, yoy_growth_pct
5	05_category_inflows.csv	144	3	month, category, net_inflow_crore
6	06_industry_folio_count.csv	21	6	month, total_folios_crore
7	07_scheme_performance.csv	40	19	amfi_code, sharpe_ratio, alpha, beta
8	08_investor_transactions.csv	32,778	13	investor_id, transaction_type, amount_inr
9	09_portfolio_holdings.csv	322	8	amfi_code, sector, weight_pct
10	10_benchmark_indices.csv	8,050	3	date, index_name, close_value

* After forward-fill for weekends and public holidays

Data Quality Summary

A comprehensive data quality assessment was performed at ingestion. All 40 AMFI scheme codes in fund_master were confirmed to exist in nav_history with no mismatches. One expected anomaly was identified: 12 null values in the yoy_growth_pct column of monthly_sip_inflows, corresponding to the first 12 months of data where no prior-year benchmark exists for year-on-year comparison.

4. ETL Pipeline & Data Architecture

4.1 ETL Pipeline Design

The ETL pipeline follows a three-stage architecture: Extract (raw data ingestion), Transform (cleaning and validation), and Load (SQLite database population). Each stage is implemented as a separate Python module in the `scripts/` directory.

Extract: Load all 10 CSV datasets using pandas. Fetch live NAV from `mfapi.in` for 6 key schemes. Inspect shape, dtypes, and null values for each dataset.

Transform: Parse dates to datetime format. Sort `nav_history` by `amfi_code` and date. Forward-fill missing NAV for weekends and holidays. Remove duplicates. Validate data ranges (NAV > 0, expense ratio 0.1%–2.5%, KYC status enum values).

Load: Execute CREATE TABLE statements from `schema.sql`. Load 10 cleaned datasets into SQLite using SQLAlchemy. Verify row counts match source CSVs. Confirm referential integrity across fact and dimension tables.

4.2 SQLite Star Schema

The database follows a star schema design with one central dimension table (`dim_fund`) and ten fact tables. All fact tables reference `dim_fund` via the `amfi_code` foreign key.

Table	Type	Rows	Primary Key	Description
<code>dim_fund</code>	Dimension	40	<code>amfi_code</code>	Fund master metadata
<code>fact_nav</code>	Fact	64,320	<code>id (auto)</code>	Daily NAV history
<code>fact_transactions</code>	Fact	32,778	<code>id (auto)</code>	Investor transactions
<code>fact_performance</code>	Fact	40	<code>id (auto)</code>	Risk-return metrics
<code>fact_aum</code>	Fact	90	<code>id (auto)</code>	AUM by fund house
<code>fact_portfolio</code>	Fact	322	<code>id (auto)</code>	Stock-level holdings
<code>fact_sip</code>	Fact	48	<code>id (auto)</code>	Monthly SIP inflows
<code>fact_category</code>	Fact	144	<code>id (auto)</code>	Category inflows
<code>fact_folio</code>	Fact	21	<code>id (auto)</code>	Industry folio count
<code>fact_benchmark</code>	Fact	8,050	<code>id (auto)</code>	Nifty 50 & Nifty 100

5. Exploratory Data Analysis

Exploratory Data Analysis was performed in notebook 03_eda_analysis.ipynb producing 10 charts exported as PNG files. The analysis covers NAV trends, AUM growth, SIP inflows, investor demographics, geographic distribution, and portfolio composition.

5.1 NAV Trend Analysis

Daily NAV for 6 bluechip funds was plotted from January 2022 to December 2025. All funds showed a strong bull run through 2023 with annualised gains of 15–28%. A mild correction was observed in early-to-mid 2024 (approximately -8% to -12% from peak), followed by recovery through Q3-Q4 2024 and continued growth into 2025.

5.2 AUM Growth

SBI Mutual Fund leads industry AUM at approximately Rs. 12.5 Lakh Crore, followed by HDFC MF at Rs. 8.2 Lakh Crore and ICICI MF at Rs. 7.8 Lakh Crore. All 10 AMCs recorded positive AUM growth across 2022–2025, reflecting broad-based growth in the Indian mutual fund industry.

5.3 SIP Inflow Trends

Monthly SIP inflows grew consistently from Rs. 11,517 Crore in January 2022 to an all-time high of Rs. 31,002 Crore in December 2025 — a 169% increase over 4 years. This growth reflects increasing adoption of systematic investing among retail investors, particularly in the 25–44 age bracket.

5.4 Investor Demographics

- Age group 25–34 represents the largest investor segment at approximately 32% of all transactions
- SIP amounts are highest in the 35–44 age bracket (avg Rs. 8,500–10,000 per transaction)
- Male investors account for approximately 62% of transactions; female investors 35%; others 3%
- Maharashtra, Delhi NCR, and Karnataka lead in total SIP investment value
- T30 cities account for approximately 68% of total investment; B30 cities 32%

5.5 Portfolio Composition

Sector allocation analysis across all equity fund holdings reveals a heavy concentration in Financial Services (approximately 28–32% weight) and Information Technology (15–20% weight). This mirrors the NIFTY 50 index composition and reflects fund manager preference for large, liquid, high-quality companies in these sectors.

5.6 Folio Count Growth

Total industry folios nearly doubled from 13.26 Crore in January 2022 to 26.12 Crore in December 2025, growing at a compound rate of approximately 18.5% annually. Equity folios drove 85% of this growth, while debt folios showed modest growth of approximately 4% annually.

6. Fund Performance Analytics

Performance analytics were computed in notebook 04_performance_analytics.ipynb for all 40 fund schemes. The analysis covers daily returns, CAGR across multiple time horizons, risk-adjusted return ratios, regression-based metrics, and maximum drawdown analysis.

6.1 Methodology

Metric	Formula	Parameters
Daily Return	$NAV(t) / NAV(t-1) - 1$	Daily frequency
CAGR	$(NAV_end / NAV_start)^{(1/n)} - 1$	n = 1, 3, 5 years
Sharpe Ratio	$(R_p - R_f) / Std(R_p) \times \sqrt{252}$	Rf = 6.5% (RBI repo rate)
Sortino Ratio	$(R_p - R_f) / Std_down \times \sqrt{252}$	Downside deviation only
Beta	$Cov(fund, bench) / Var(bench)$	Nifty 100 benchmark
Alpha	Intercept x 252	OLS regression, annualised
Max Drawdown	$min(NAV / running_max - 1)$	Full history
Fund Score	30% CAGR + 25% Sharpe + 20% Alpha + 15% Expense(inv) + 10% DD(inv)	Composite 0-100

6.2 Key Performance Findings

- Large Cap Direct funds consistently deliver the best risk-adjusted returns with Sharpe ratios above 1.0
- The average 3-year CAGR across all 40 funds is approximately 14.8%, outperforming fixed deposit rates
- Small Cap funds show the highest absolute returns (3yr CAGR 18–22%) but also highest volatility
- Direct plans outperform Regular plans by 0.5–1.2% annually due to lower expense ratios
- Funds with expense ratio below 1% (Direct plans) show significantly better net returns over 5 years
- Maximum drawdown ranged from -8% to -28% across funds, with the worst during the 2024 correction

6.3 Fund Scorecard — Top 10 Funds

Rank	Fund Name	3yr CAGR%	Sharpe	Alpha%	Exp Ratio%	Score
1	SBI Bluechip Direct	16.8	1.24	2.1	0.45	82.4
2	HDFC Top 100 Direct	15.9	1.18	1.8	0.52	79.6
3	ICICI Pru Bluechip Direct	16.2	1.15	1.6	0.48	78.3
4	Axis Bluechip Direct	15.4	1.12	1.5	0.43	76.9
5	Kotak Bluechip Direct	15.1	1.09	1.3	0.46	75.2

7. Advanced Analytics & Risk Metrics

Advanced analytics were implemented in notebook 05_advanced_analytics.ipynb, covering Value at Risk, rolling performance metrics, investor behaviour analysis, and portfolio concentration measures.

7.1 Historical Value at Risk (VaR 95%) and CVaR

Historical VaR at 95% confidence was computed as the 5th percentile of the daily return distribution for each fund. CVaR (Conditional VaR, also known as Expected Shortfall) was computed as the mean of all returns falling below the VaR threshold, providing a measure of tail risk beyond the VaR level.

- Small Cap funds exhibit the highest VaR (worst-case daily loss of -2.1% to -2.8%)
- Large Cap Direct funds show the lowest VaR (-1.1% to -1.4%), confirming defensive characteristics
- CVaR exceeds VaR by 25-40% on average, indicating significant tail risk on worst-case days
- Funds with high Sharpe ratios tend to have lower VaR, confirming risk-return trade-off

7.2 Rolling 90-Day Sharpe Ratio

Rolling Sharpe ratios were computed using a 90-day rolling window, allowing analysis of how risk-adjusted performance evolved over time. The 2023 bull market produced exceptional rolling Sharpe values above 2.0 for most bluechip funds. The 2024 correction compressed rolling Sharpe below 1.0 before recovery in Q4 2024.

7.3 Investor Cohort Analysis

Investors were grouped by their first transaction year (2022, 2023, 2024, 2025) to analyse cohort behaviour. The 2022 cohort (earliest adopters) shows the highest average SIP amounts, while the 2024-2025 cohorts have the most investors but lower average ticket sizes, reflecting the growing democratisation of mutual fund investing among retail participants.

7.4 SIP Continuity Analysis

For investors with 6 or more SIP transactions, the average gap between consecutive SIP dates was computed. Investors with an average gap exceeding 35 days were flagged as at-risk of SIP discontinuity. Approximately 15–20% of frequent SIP investors are classified as at-risk, representing a significant churn prevention opportunity.

7.5 Sector HHI Concentration

The Herfindahl-Hirschman Index ($HHI = \text{sum of squared sector weights}$) was computed for each equity fund to measure portfolio concentration. Funds with HHI above 0.15 are classified as highly concentrated, typically with 2–3 dominant sectors. Funds with HHI below 0.08 offer better sector diversification and are generally more suitable for conservative investors.

8. Dashboard Overview

A 4-page interactive dashboard was developed using Tableau Public (bluestock_mf_dashboard.twbx). Each page focuses on a distinct analytical theme with interactive slicers and drill-through capabilities.

Page 1 — Industry Overview

- KPI cards: Total AUM, SIP Inflows, Total Folios, Scheme Count
- Line chart: Industry AUM trend 2022–2025
- Horizontal bar chart: AUM by AMC (latest month)

Page 2 — Fund Performance

- Scatter plot: Return vs Risk (bubble size = AUM)
- Sortable fund scorecard table with all 40 schemes
- Slicers: Fund House, Category, Plan

Page 3 — Investor Analytics

- Bar chart: Transaction amount by state
- Donut chart: SIP vs Lumpsum vs Redemption split
- Bar chart: Average SIP amount by age group
- Line chart: Monthly transaction volume

Page 4 — SIP & Market Trends

- Dual-axis: SIP inflow (bar) + Nifty 50 (line) 2022–2025
- Heatmap: Category inflows by month
- Bar chart: Top 5 categories by net inflow FY25

9. Key Findings & Insights

1. SBI Dominates AUM

SBI Mutual Fund leads industry AUM at Rs. 12.5 Lakh Crore, representing approximately 15% of total industry AUM.

2. SIP Growth Momentum

Monthly SIP inflows grew 169% over 4 years, hitting an all-time high of Rs. 31,002 Crore in December 2025, demonstrating robust retail participation.

3. Folio Doubling

Total industry folios doubled from 13.26 Crore to 26.12 Crore (2022–2025), indicating successful financial inclusion and investor outreach.

4. Large Cap Superiority

Large Cap Direct funds deliver the best risk-adjusted returns with Sharpe ratios above 1.0, making them the optimal choice for most retail investors.

5. Youth-Led Investing

The 25–34 age group is the largest investor segment, indicating strong uptake of digital-first mutual fund platforms among young professionals.

6. Geographic Concentration

Top 3 states (Maharashtra, Delhi, Karnataka) account for approximately 45% of total SIP investment, highlighting the urban concentration of formal investing.

7. Sector Concentration Risk

Financial Services and IT together account for 43–52% of equity fund portfolios, creating meaningful sector concentration risk across the industry.

8. SIP Continuity Risk

15–20% of active SIP investors are at risk of discontinuation (avg gap > 35 days), representing a significant retention opportunity for fund houses.

9. Direct Plan Advantage

Direct plans outperform Regular plans by 0.5–1.2% annually — over 10 years, this compounding advantage translates to 8–14% additional wealth creation.

10. Bull Run of 2023

All 40 fund schemes showed strong NAV growth in 2023, with bluechip funds delivering 18–26% returns, the strongest single-year performance in the dataset.

10. Limitations & Recommendations

10.1 Limitations

- The dataset covers 40 out of approximately 1,500+ AMFI-registered schemes; findings may not generalise to all categories
- CAGR calculations use calendar-day denominators in the current implementation; trading-day denominator (252 days) would be more precise
- The fund recommender is rule-based (Sharpe ranking within risk grade); a machine learning model would produce more personalised recommendations
- Benchmark comparison uses Nifty 100 as a proxy; individual funds may be more accurately benchmarked against their specific SEBI-mandated benchmark
- Dashboard data is static (point-in-time CSVs); a production system would connect to live AMFI data feeds
- Investor demographic data may not fully represent the universe of mutual fund investors in India

10.2 Recommendations

- Implement a live data pipeline using the AMFI API or CDSL data feeds for real-time NAV and portfolio updates
- Expand coverage to all SEBI mutual fund categories including ELSS, Index Funds, ETFs, and International Funds
- Build an ML-based fund recommender using collaborative filtering on investor transaction history
- Integrate macro-economic indicators (inflation, RBI policy rates, GDP growth) for contextual performance analysis
- Develop an automated weekly email report for fund performance summaries using HTML templates
- Implement portfolio optimisation using the Markowitz Efficient Frontier for selected fund universes
- Add scenario analysis and Monte Carlo simulation for 5-year NAV growth projections
- Build SIP continuity alerts for at-risk investors to reduce churn and improve long-term investor outcomes

End of Report

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